Recap: Lecture 23

In Lecture 23, we

- Applied SoV to a second problem involving the 1D heat equation and homogeneous BCs
- Solved a problem involving inhomogeneous BCs by writing

$$u(x,t) = u_S(x) + u_T(x,t)$$

  where u_S satisfies the inhomogeneous BCs and is time independent and u_T satisfies homogeneous BCs

Lecture 24

In Lecture 24, we will:

1. Consider 2D heat equation on circular panels
2. Apply SoV to an axisymmetric problem with homogeneous BCs
3. Find the steady state solution for a problem involving inhomogeneous BCs by applying SoV to Laplace's equation

https://upload.wikimedia.org/wikipedia/commons/5/59/SolarStirlingEngine.jpg

https://www.researchgate.net/figure/US-systems-22-kWel-SAIC-and-10-kWel-WGA_fig11_282859077

Case study 8: Heat conduction in 2D

Heat conduction in 2D: Problem 1

Find the evolution of the temperature of circular panel whose edge is held at zero temperature and subject to an axisymmetric initial condition.

- Laplace's equation in cylindrical polar coordinates for $u(\mathbf{x}, t) = u(r, \theta, t)$ is

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u \equiv \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} \quad \text{for} \quad 0 < r < a, \;\; 0 \leqslant \theta < 2\pi, \;\; t > 0$$

- Boundary condition

$$u(r = a, \theta, t) = 0 \quad \text{for} \quad 0 \leqslant \theta < 2\,\pi, \;\; t > 0$$

- Initial condition

$$u(r, \theta, t = 0) = f(r)$$

$$\boxed{\text{Seek solution using separation of variables}}$$

1. Separate variables to turn single PDE into multiple ODEs

2. Apply the boundary condition to find the eigenvalues and eigenfunctions

3. Solve the remaining ODEs for each eigenvalue

4. Apply puperposition Principle to get a series general solution

5. Apply the initial conditions to find the series coefficients

Laplace's equation: Steady-state solutions of 2D heat equation

- The heat equation in 2D is
$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$
- Decomposed the solution as
$$u(\mathbf{x}, t) = u_T(\mathbf{x}, t) + u_S(\mathbf{x}) \quad \text{such that} \quad u(\mathbf{x}, t) \to u_S(\mathbf{x}) \quad \text{as } t \to \infty$$
- Set
$$\frac{\partial u_T}{\partial t} = \alpha \nabla^2 u_T \quad \Rightarrow \quad \nabla^2 u_S = 0$$
  i.e., the steady-state solution satisfies Laplace's equation
- We use the steady-state solution to satisfy inhomogeneous BCs

Heat conduction in 2D: Problem 2

Find the steady state for a circular panel, subject to an inhomogeneous BC.

- Laplace's equation in cylindrical polar coordinates for $u_S(\mathbf{x}) = u_S(r, \theta)$ is

$$\nabla^2 u_S \equiv \frac{\partial^2 u_S}{\partial r^2} + \frac{1}{r}\frac{\partial u_S}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u_S}{\partial \theta^2} = 0 \quad \text{for} \quad 0 < r < a, \;\; 0 \leqslant \theta < 2\pi$$

- Inhomogeneous BC

$$u_S(r = a, \theta, t) = f(\theta) \quad \text{for} \quad 0 \leqslant \theta < 2\,\pi$$

Seek solution using <u>an adapted version</u> of separation of variables

1. Separate variables to turn single PDE into multiple ODEs

2. Find eigenvalues and eigenfunctions

3. Solve the remaining ODEs for each eigenvalue

4. Apply Superposition Principle to get a series general solution

5. Apply the inhomogeneous (boundary) conditions to find the series coefficients